more than 120˚C. When the bearing is cooled down after mounting, the inner ring contracts in the axial direction, creating a clearance between the inner ring and shoulder. To prevent this clearance from occurring, the inner ring and shoulder should be kept in tight contact with each other using a nut. The relationship between heat temperature differences, bearing dimensions and the amount of inner ring expansion is shown in **Fig. 8.2**.

(3) Driving Fits

When interference is low in bearings with small dimensions, driving fitting is sometimes used during mounting. In this method, a hammer (made of resin) is used to drive the ring in. The mounting dolly is placed on the ring's surface, and force must be applied evenly to its entire surface. However, this method is not generally recommended because it often causes bearing damage.

8.3 Insertion and Removal Force

Insertion force, which is the force to press the bearing onto a shaft or into a housing with interference, and removal force, which is the force to remove the bearing from a shaft or housing, are calculated using the following equation.

$$K = \mu p_m \pi d \text{ (or } D) \cdot B \quad \text{.....(8.1)}$$

where, K: Insertion force or removal force (N), {kgf}
p_m: Surface pressure on the fitting surface (refer to **Table 5.1**) (N/mm^2), {kgf/mm^2}
d: Nominal bore diameter (mm)
D: Nominal outside diameter (mm)
B: Nominal inner (outer) ring width (mm)
μ : Sliding friction coefficient of shaft and inner ring, or of housing and outer ring (from **Table 8.1**)

The value p_m in equation **8.1** is calculated from the value found in **Table 5.1**. However, in the case of an inner ring and shaft, the value k

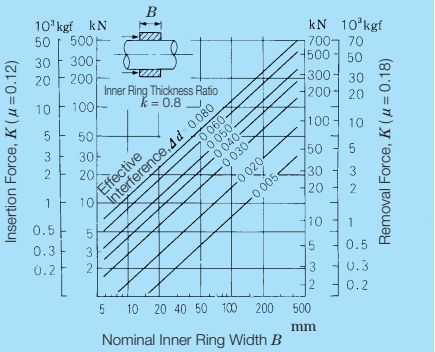


Fig. 8.3
Insertion and Removal Force for Inner Ring

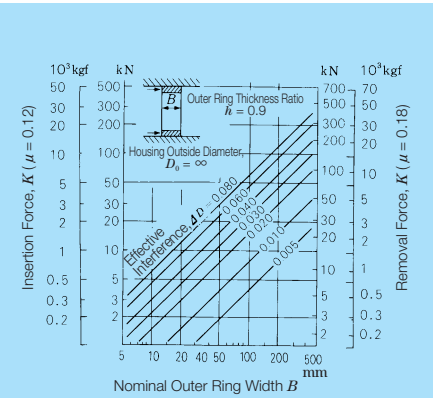


Fig. 8.4
Insertion and Removal Force for Outer Ring

in **Table 5.1** varies, depending on bearing type and dimension series, between 0.7 to 0.9. Assuming $k = 0.8$, the line graphs shown in **Fig. 8.3** are calculated when the shaft is solid (i.e., $k_0 = 0$).

Likewise, the insertion and removal forces calculated for outer rings and the housings at $h = 0.9$ and $h_0 \fallingdotseq 0$ are shown in **Fig. 8.4**.

The actual insertion and removal force may significantly exceed calculated values due to mounting error. The values in **Fig. 8.3** and **8.4** are considered only as guide values. When designing withdrawal tools, rigidity tolerances should be 5 to 6 times the values found in the figures.

Table 8.1 Coefficient of Sliding Friction for Insertion and Removal

	μ
When inner ring is pressed onto cylindrical shaft or outer ring is pressed into housing bore	0.12
When inner ring is removed from cylindrical shaft or outer ring is removed from housing bore	0.18

8.4 Operating Inspection

After mounting has been completed, a running test should be conducted to determine if the bearing has been properly mounted. Any sharp increase in speed to normal levels should be avoided as this could cause damage in the bearing in the event of improper mounting or heat seizure of the bearing in the event of incorrect lubrication.

During the operating inspection, it is necessary to check for the following items to determine if there are any abnormalities:

(1) Manual Inspection
- Sticking (dust and dirt, flaws)
- Uneven torque (improper mounting)
- Excessive torque (inadequate clearance, high seal friction)

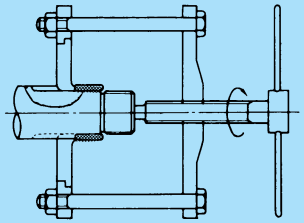


Fig. 8.5
Dismounting of Inner Ring Using Withdrawal Tool

(2) Powered Operating Inspection
- Abnormal noise (dust and dirt, flaws, incorrect lubrication, excessive or insufficient residual clearance)
- Operating temperature saturation, temperature rise per unit hours (incorrect lubrication, improper mounting)

During the powered inspection, gradually increase speed while ensuring that no abnormalities appear at each stage. Only the powered operating inspection is necessary for large bearings.

8.5 Dismounting

Bearings are dismounted for periodic inspections and other purposes. When the bearing is planned to be reused after dismounting or to have its condition examined, dismounting has to be carried out with extra care. When the fit is tight, dismounting is difficult. Therefore, careful consideration should be taken in designing the structure of the bearing vicinity.

When dismounting bearings, specially designed withdrawal tools (**Fig. 8.5**) or presses (**Fig. 8.6**) are to be used.

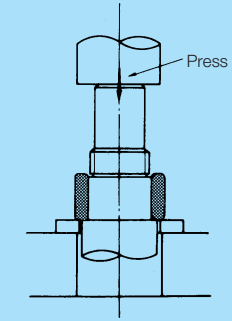


Fig. 8.6
Dismounting of Inner Ring by Press

8.6 Cleaning

When a dismounted bearing is being reused, the bearing should be cleaned to remove harmful foreign matter such as sand metallic particles, sludge and deteriorated lubricant. Light oil or kerosene is generally used as a cleaning agent. The cleaning procedure should be divided into a rough wash and a finishing wash, and a container with an elevated metal net should be used so that the bearing does not contact the bottom of the container.

Maintaining the cleanliness of the cleaning oil is very important. It should therefore be circulated through a filter at all times.

BEARING TABLES



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CAGE & NEEDLE ROLLER ASSEMBLIES

Pages

Metric Design Cage & Needle Roller Assemblies

Inscribed Circle Diameter 4–150 mm B12–B29

Inch Design Cage & Needle Roller Assemblies

Inscribed Circle Diameter 9.525–127.000 mm B30–B33

Cage & Needle Roller Assemblies for Large Ends of Connecting Rods

Inscribed Circle Diameter 12–30 mm B34

Cage & Needle Roller Assemblies for Small Ends of Connecting Rods

Inscribed Circle Diameter 9–19 mm B35

NSK cage & Needle roller assemblies consist of a complement of needle rollers held in place by a cage. Their unitized design makes for easy handling and installation. With no inner or outer ring, the low cross-section provides maximum load-carrying capability within the smallest envelope. Properly lubricated, they can operate at high speeds thanks to the unique cage, which guides the rollers precisely. The controlled contour rollers in cage & Needle roller assemblies have an optimum profile which reduces end stresses, allows operation under moderate misalignment and prolongs bearing life.

DESIGN AND TYPES

The most common cage used in cage & Needle roller assemblies is the M-type. M-type cages are either pressed from steel plate or machined. These cages undergo surface hardening after processing to attain high wear resistance and rigidity. The cages facilitate optimum lubrication conditions because lubricant can be easily replenished. Other cages include those made by precision pressing, welded cages and polyamide molded cages. Welded cages, which are produced by welding pressed steel, as especially suited for mass production. Applying its considerable experience and knowledge, NSK supplies cage & Needle roller assemblies of the highest quality.

Codes for the various types of cage & Needle roller assemblies are shown in **Table 1**. There are many types of cage & Needle roller assemblies other than those in **Table 1**, including split cages. Cage & Needle roller assemblies for engine connecting rods are included on page B9 as one of a special series.

Table 1 Cage & Needle Roller Assembly Codes

Cage Type		Code	
		Metric	Inch
Single Row	M Cage	WFJ, FWF	WJ
	FBN Cage	FBN	—
	Welded Cage	FWJC, FWJW	WJC
	Polymide Cage	FBNP	—
Double Row	M Cage	FWF ... W, FDJ	DJ
	FBN Cage	FBN ... W	—
	Welded Cage	FDJC	DJC
	Polymide Cage	FBNP ... W	—

An example of a bearing number is as follows:

(Example) FWD-202620

- Width (mm)
- Circumscribed Circle Diameter (mm)
- Inscribed Circle Diameter (mm)
- Code

For inch bearings, each dimension is expressed with two digit whole numbers in units of 1/16 of an inch following the type code.

ACCURACY STANDARDS

NSK cage & Needle roller assemblies use rollers classified as JIS third class or higher. Roller accuracy is shown in **Table 2**.

Table 2 Accuracy of Needle Rollers

Units: μm

Class	Tolerance in Average Value of Diameter high	Tolerance in Average Value of Diameter low	Diameter Variation in a Single Radial Plane, (1) VD_{WP} max	Deviation from Circular Form, (1) ΔR max	Diameter Variation within Gauge Lot, (1) VD_{WL} max	Deviation in length, (2) ΔL_{WS}
2	0	−5	1	1	2	h13
3	0	−5	1.5	1.5	3	h13
5	0	−5	2	2.5	5	h13

Note (1) Applies at the center of roller's length.
(2) Depends on L_w classification. h13 is shown in **Table 5**, page B201.

Remarks Deviation between the actual measured diameter at any point over the entire length of roller and the actual maximum diameter at the center of the entire length must not exceed the following values:
Class 2: 0.5 μm
Class 3: 0.8 μm
Class 5: 1 μm

APPLICATION

Mounting

(1) With cage & Needle roller assemblies, the shaft and housing are generally used as raceway rings. It is important to satisfy the finish accuracy and hardness requirements in **Table 3**.

Table 3 Accuracy and Hardness for Shaft and Housing

Item	Shaft	Housing
Out-of-Roundness Tolerance	$\frac{IT3}{2}$	$\frac{IT3}{2}$
Cylindricity Tolerance	$\frac{IT3}{2}$	$\frac{IT3}{2}$
Roughness, Ra	0.4	0.8
Hardness	HRC58 to 64 Appropriate depth of hardened layer required	HRC58 to 64 Appropriate depth of hardened layer required

Remarks For details, please refer to Section **5.2** on page A34 and Section **5.3** on page A35.

(2) Guide width for cage & Needle roller assemblies
The tolerances for the guide width, b, are based on cage width, B_C, and shown in **Table 4** (see **Fig. 1**).

Table 4 Tolerances for Guide Width, b

Type	Tolerance for b
Metric	$B_C{}^{+0.2}_{0}$
Inch	$B_C{}^{+0.4}_{+0.2}$

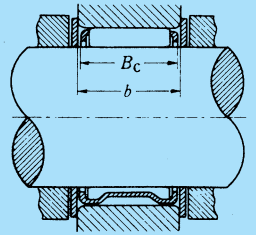


Fig. 1

(3) Fillet Radius
The fillet radius of the shaft and housing shoulder in contact with the cage end surface should not interfere with the cage itself.

Fitting Tolerances for Shafts and Housing Bores

Fitting tolerances for shafts and housing bores under various operating conditions are shown in **Table 5**.

Table 5 Fitting Tolerances for Shafts and Housing Bores

Operating Conditions		High Accuracy, Oscillating Motion	Normal	High Temperature, Large Shaft Deflection, Mounting Error
Fitting Tolerance for Housing Bore		G6	G6	G6
Fitting Tolerance for Shaft	$F_w \leqq 50$mm	j5	h5	f6
	$F_w > 50$mm	h5	g5	

Shaft Inclination

Permissible shaft inclinations from elastic deflection by external force and mounting error are shown in **Table 6**. When shaft inclination exceeds that in this table, please consult NSK because it is necessary to change the amount of the rollers' crowning and increase the internal clearance.

Table 6 Shaft Inclination

Cage Width (mm)	Permissible Inclination (mm/mm) max
$\leq$ 25	0.0015
(25) to 50	0.0010
$>$ 50	0.0005

Cage & Needle Roller Assemblies for Connecting Rods

Cage & Needle roller assemblies for connecting rods used in small general-purpose engines such as those in compact cars, motorcycles, outboard marine motors, snow-mobiles and chain-saws operate in harsh conditions characterized by shock loads, complex motion, high temperatures, and lubrication difficulties. Cage & Needle roller assemblies FWF ... E, FBN ... E are specially designed for such conditions. Cage & Needle roller assemblies for connecting rods contain a highly rigid and wear-resistant lightweight cage and high-accuracy rollers with special crowning.

TYPES

Cage & Needle Roller Assemblies for the Large Ends of Connecting Rods FWF ... E

FWF ... E Cage & Needle roller assemblies for large ends of connecting rods are lightweight and highly rigid. Their outside surface is accurately finished to maintain proper guide clearance during operation. Special steel that is hardened to the proper depth by carbonitriding is used as material for the cages. Depending on operating conditions, wear resistance is enhanced by surface treatment with non-ferrous metal.

For integrated one-piece crankshafts, split cage & Needle roller assemblies are also manufactured.

Cage & Needle Roller Assemblies for the Small Ends of Connecting Rods FBN ... E

Owing to their high static load rating, cage & Needle roller assemblies for small ends of connecting rods contain many long rollers with diameters of 1.5 to 2.5 mm, lowering the contact pressure within the load zone. Depending on the structure of the engine, a bore surface-guided or outside diameter-guided cage is used.

APPLICATIONS

Specifications for Connecting Rods, Crank Pins and Gudgeon Pins

The small and large end bores of the connecting rod, which function as the raceway surface of the cage & Needle roller assemblies, and the outside diameter of the crank pin and gudgeon pin must have the following specifications in order to obtain the beat performance, even under harsh operating conditions, from cage & Needle roller assemblies for connecting rods.

(1) Raceway Surface Roughness

The roughness of the raceway surface of the crank pin and gudgeon pin, which can lead to premature wear under severe operating conditions, should be approximately 0.1μmR_{max}. Roughness for the large end bore and small end bore of the connecting rod should be 0.25μmRa or less.

(2) Raceway Surface Hardness and Effective Hardened Layer Depth

The surface hardness of the raceway should be HRC60 to 64. The effective hardened layer depth (depth to HV550) varies depending on load conditions, but should range from 0.6 to 1.2 mm.

(3) Raceway Accuracy

Requirements for out-of-roundness and cylindricity of the raceway are listed in **Table 1**.

Table 1 Out-of-Roundness and Cylindricity of Raceway

Units: μm

Diameter Classification (mm)		Large/Small End Bore		Crank Pin/Gudgeon Pin	
over	incl	Out-of-roundness max	Cylindricity max	Out-of-roundness max	Cylindricity max
—	18	5	3	4	2
18	30	6	4	4	2
30	40	8	4	5	3

(4) Parallelism for Large and Small End Bores of Connecting Rods

Improper parallelism between the large end bore and small end bore of the connecting rod decreases engine performance. The two bores therefore have to be finished with a high degree of accuracy. It is recommended that parallelism is less than or equal to 0.04 mm for each 100 mm for normal and low speeds, and less than or equal to 0.02 mm for each 100 mm for high speeds.

(5) Abutment Tolerance for Mounting

Radial internal clearance in cage & Needle roller assemblies for connecting rod large and small ends should be determined by engine type, characteristics, and operating conditions. In general, h5 for the piston diameter, G6 for the samall end bore, g5 or f5 for the crank pin diameter, and G6 for the large end bore should be applied. When the radial internal clearance has to be strictly limited, it is recommended to use the values in **Table 2**. As to the large end, the upper limit of the internal clearance should be the target value under high-speed rotation.

Table 2 Recommended Radial Internal Clearance

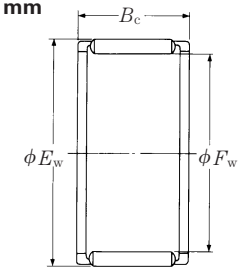
Units: μm

Shaft Diameter Classification (mm)		Large End		Small End	
over	incl	min	max	min	max
6	10	—	—	5	15
10	18	12	24	6	18
18	30	14	28	7	20
30	—	18	34	—	—

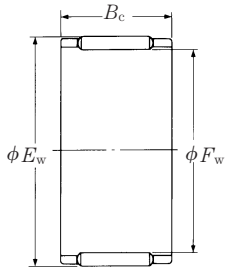
FWF

FBN • FBNP

Inscribed Circle Diameter 4 — 12 mm



FWF



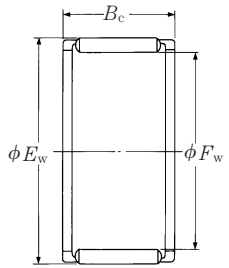
FBN, FBNP

Boundary Dimensions (mm)			Basic Load Ratings (N)		{kgf}		Limiting Speeds (min^{-1})	Bearing Numbers	Mass (g)
F_w	E_w	B_c $^{-0.2}_{-0.55}$	C_r	C_{or}	C_r	C_{or}	Oil		approx
4	7	7	1 710	1 220	175	124	120 000	＊**FBNP-477**	0.5
5	8	8	2 330	1 860	237	189	95 000	＊**FBNP-588**	1.0
6	9	8	2 200	1 780	224	182	75 000	＊**FBNP-698**	1.2
	9	10	2 980	2 630	305	268	75 000	＊**FBNP-6910**	1.5
	10	13	4 300	3 600	435	365	80 000	＊**FBNP-61013**	2.8
7	10	8	2 840	2 560	290	261	67 000	＊**FBNP-7108**	1.3
	10	10	3 650	3 550	375	360	67 000	＊**FBNP-71010**	1.6
8	11	8	3 050	2 900	310	296	56 000	＊**FBNP-8118**	1.4
	11	10	3 500	3 450	360	355	56 000	＊**FBNP-81110**	1.8
	11	13	4 750	5 150	485	525	56 000	＊**FBNP-81113**	2.6
9	12	10	3 750	3 850	380	395	50 000	＊**FBNP-91210**	2.0
	12	13	5 100	5 750	520	585	50 000	＊**FBNP-91213**	2.6
10	13	8	3 450	3 600	355	365	45 000	**FBN-10138**	2.0
	13	10	3 950	4 300	405	435	45 000	**FBN-101310**	2.2
	13	11	4 450	5 000	455	505	45 000	**FBN-101311**	2.4
	13	13	5 400	6 350	550	650	45 000	**FBN-101313**	2.9
	14	8	3 750	3 300	385	340	45 000	**FBN-10148**	2.5
	14	10	5 350	5 300	545	540	45 000	**FWF-101410**	3.1
	14	13	6 500	6 750	660	690	45 000	**FWF-101413**	4.0
	16	12	7 700	6 650	785	680	48 000	**FWF-101612**	6.0
11	14	10	3 950	4 350	400	440	40 000	**FBN-111410**	2.6
12	15	9	4 350	5 100	445	520	36 000	**FBN-12159**	2.4
	15	10	4 350	5 100	445	520	36 000	**FBN-121510**	2.6
	15	12	5 400	6 750	555	690	36 000	**FBN-121512**	3.1
	15	13	5 950	7 600	605	775	36 000	**FBN-121513**	3.4
	16	10	5 700	5 950	580	610	38 000	**FWF-121610**	3.6
	16	13	7 350	8 350	750	850	38 000	**FWF-121613**	4.6
	17	13	8 250	8 350	840	855	38 000	**FWF-121713**	6.0
	18	12	9 150	8 650	935	885	38 000	**FWF-121812**	6.9

＊ These bearings have polyamide cages. The maximum permissible operating temperature for these bearings is 100℃ for continued operation and 120℃ for short periods.

FWF • FWJ

Inscribed Circle Diameter 14 — 17 mm



FWF, FWJ

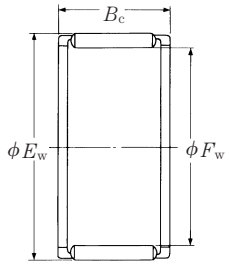
Boundary Dimensions (mm)			Basic Load Ratings (N)		Basic Load Ratings {kgf}		Limiting Speeds (min^{-1})	Bearing Numbers	Mass (g)
F_w	E_w	B_c $^{-0.2}_{-0.55}$	C_r	C_{or}	C_r	C_{or}	Oil		approx
14	18	10	6 750	7 750	690	790	32 000	**FWF-141810**	4.1
	18	11	6 650	7 600	675	775	32 000	**FWF-141811**	4.5
	18	13	8 050	9 750	820	995	32 000	**FWF-141813**	5.3
	18	17	10 700	14 100	1 090	1 430	32 000	**FWF-141817**	6.9
	18	20	12 100	16 400	1 230	1 670	32 000	**FWF-141820**	10
	20	12	9 850	9 800	1 010	1 000	32 000	**FWF-142012**	7.8
	20	17	13 400	14 600	1 370	1 490	32 000	**FWF-142017**	11
15	19	10	7 050	8 400	720	855	28 000	**FWF-151910**	4.3
	19	13	8 400	10 500	860	1 070	28 000	**FWF-151913**	5.6
	19	17	11 200	15 200	1 140	1 550	28 000	**FWF-151917**	7.4
	20	13	9 950	11 300	1 010	1 150	30 000	**FWF-152013**	7.2
	21	15	12 600	13 600	1 280	1 380	30 000	**FWF-152115**	10
	21	17	13 400	14 800	1 370	1 510	30 000	**FWF-152117**	12
	21	21	16 300	18 900	1 660	1 930	30 000	**FWF-152121A**	14
16	20	10	7 350	9 000	750	920	26 000	**FWF-162010**	4.6
	20	13	8 800	11 300	895	1 150	26 000	**FWF-162013**	6.0
	20	17	11 700	16 300	1 190	1 660	26 000	**FWF-162017**	7.8
	21	13	10 500	12 300	1 070	1 250	28 000	**FWF-162113**	7.7
	21	17	13 000	16 200	1 330	1 660	28 000	**FWF-162117**	10
	21	26	20 000	28 200	2 040	2 880	28 000	**FWF-162126**	15
	21	27	20 800	29 600	2 120	3 000	28 000	**FWF-162127**	16
	22	12	10 500	10 900	1 070	1 120	28 000	**FWF-162212**	8.7
	22	13	10 900	11 500	1 120	1 180	28 000	**FWF-162213**	9.4
	22	17	14 700	16 900	1 500	1 720	28 000	**FWF-162217**	12
	22	20	17 000	20 300	1 730	2 070	28 000	**FWF-162220**	14
	22	22	18 700	22 900	1 900	2 340	28 000	**FWF-162222**	16
	22	24	20 300	25 600	2 070	2 610	28 000	**FWF-162224**	17
	24	20	20 800	22 200	2 120	2 270	30 000	**FWF-162420**	20
17	21	10	7 650	9 650	780	985	26 000	**FWF-172110**	4.8
	21	13	10 200	14 000	1 040	1 420	26 000	**FWJ-172113**	6.3
	21	15	10 800	15 000	1 100	1 530	26 000	**FWF-172115**	7.5
	21	16	11 500	16 300	1 170	1 670	26 000	**FWF-172116**	8.0
	21	17	11 500	16 300	1 170	1 670	26 000	**FWF-172117**	8.2
	22	20	15 400	20 400	1 570	2 080	26 000	**FWF-172220**	16
	23	17	15 100	17 800	1 540	1 810	26 000	**FWF-172317**	14

FWF • FWJ

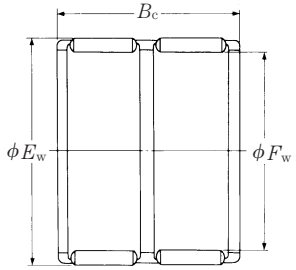
FWF···W

FBN···W

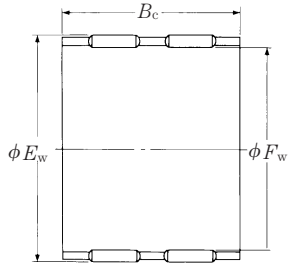
Inscribed Circle
Diameter 17.57 — 21 mm



FWF, FWJ



FWF···W



FBN···W

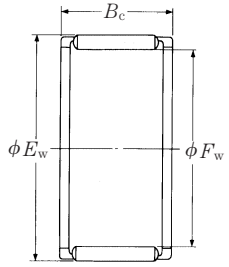
Boundary Dimensions (mm)			Basic Load Ratings (N)		Basic Load Ratings {kgf}		Limiting Speeds (min^{-1})	Bearing Numbers	Mass (g)
F_w	E_w	B_c $^{-0.2}_{-0.55}$	C_r	C_{or}	C_r	C_{or}	Oil		approx
17.57	22.6	21	16 600	22 700	1 700	2 320	24 000	**FWF-172221Z**	13
	23.6	32	29 300	42 000	2 990	4 300	26 000	**FWJ-172332Z**	25
18	22	10	7 900	10 300	805	1 050	24 000	**FWF-182210**	5.1
	22	13	9 450	12 900	965	1 310	24 000	**FWF-182213**	6.6
	22	17	12 600	18 600	1 280	1 900	24 000	**FWF-182217**	8.7
	22	28	16 100	25 400	1 640	2 590	24 000	**FBN-182228W**	16
	23	20	16 600	22 900	1 690	2 340	24 000	**FWF-182320**	13
	24	12	11 700	13 000	1 190	1 320	24 000	**FWF-182412**	9.6
	24	13	12 800	14 600	1 300	1 490	24 000	**FWF-182413**	10
	24	17	17 400	21 600	1 770	2 210	24 000	**FWJ-182417**	14
	24	20	18 900	24 100	1 930	2 460	24 000	**FWF-182420**	16
	25	14	14 200	15 200	1 450	1 550	24 000	**FWF-182514**	13
	25	17	16 500	18 300	1 680	1 860	24 000	**FWF-182517**	19
	25	22	21 300	25 500	2 170	2 600	24 000	**FWF-182522**	21
19	23	13	9 900	13 900	1 010	1 420	22 000	**FWF-192313**	7.0
	23	17	12 300	18 500	1 260	1 880	22 000	**FWF-192317**	9.1
	24	28	23 800	37 000	2 420	3 750	22 000	**FWJ-192428**	19
	31	24	31 500	31 500	3 200	3 200	26 000	**FWF-193124**	45
19.1	31.1	24	31 000	30 500	3 150	3 150	24 000	**FWJ-193124Z**	46
20	24	10	8 000	10 700	815	1 090	20 000	**FWF-202410**	5.6
	24	13	9 700	13 700	990	1 400	20 000	**FWF-202413**	7.3
	24	17	12 400	18 800	1 260	1 910	20 000	**FWF-202417**	9.5
	25	40	30 000	50 500	3 050	5 150	22 000	**FWF-202540W**	29
	26	12	12 200	14 100	1 250	1 440	22 000	**FWF-202612**	10
	26	13	13 100	15 500	1 340	1 580	22 000	**FWF-202613**	11
	26	16	16 600	21 000	1 700	2 140	22 000	**FWF-202616**	14
	26	17	18 000	23 200	1 830	2 370	22 000	**FWJ-202617**	15
	26	20	19 200	25 200	1 960	2 570	22 000	**FWF-202620**	18
	26	24	23 200	32 000	2 360	3 250	22 000	**FWF-202624**	21
	28	20	19 800	21 700	2 020	2 210	22 000	**FWF-202820**	28
21	25	13	10 500	15 500	1 070	1 580	20 000	**FWF-212513**	7.6
	25	17	13 100	20 600	1 340	2 100	20 000	**FWF-212517**	10

FWF • FWJ

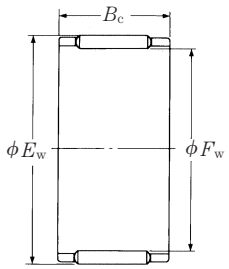
FBN

FBN···W

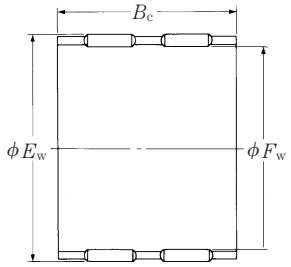
Inscribed Circle
Diameter 22 — 26 mm



FWF, FWJ



FBN



FBN···W

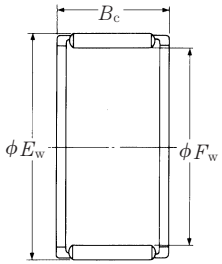
Boundary Dimensions (mm)			Basic Load Ratings (N)		Basic Load Ratings {kgf}		Limiting Speeds (min^{-1})	Bearing Numbers	Mass (g)
F_w	E_w	B_c $^{-0.2}_{-0.55}$	C_r	C_{or}	C_r	C_{or}	Oil		approx
22	26	10	8 600	12 200	880	1 240	19 000	**FWF-222610**	6.1
	26	11	8 600	12 200	880	1 240	19 000	**FWF-222611**	7.2
	26	13	10 300	15 300	1 050	1 560	19 000	**FWF-222613**	7.9
	26	17	13 000	20 700	1 330	2 110	19 000	**FWF-222617**	10
	27	20	18 000	27 000	1 840	2 760	20 000	**FWF-222720**	16
	28	13	13 700	16 700	1 390	1 700	20 000	**FWF-222813**	12
	28	17	17 300	22 700	1 760	2 310	20 000	**FWF-222817**	16
	28	23	23 600	34 000	2 410	3 450	20 000	**FWF-222823**	22
	29	16	17 100	20 200	1 750	2 060	20 000	**FWF-222916**	18
	30	14	17 400	19 000	1 770	1 930	20 000	**FWF-223014**	24
23	28	24	22 400	36 000	2 280	3 700	19 000	**FWF-232824**	20
24	28	13	10 500	16 100	1 070	1 650	17 000	**FWF-242813**	8.6
	28	17	14 600	24 700	1 490	2 510	17 000	**FWF-242817**	11
	28	25	15 900	27 400	1 620	2 800	17 000	**FBN-242825W**	19
	29	13	10 700	14 100	1 090	1 430	18 000	**FBN-242913**	11
	30	17	17 900	24 300	1 830	2 480	18 000	**FWF-243017**	14
25	29	10	9 350	14 100	950	1 440	17 000	**FWF-252910**	6.9
	29	13	11 300	18 000	1 150	1 830	17 000	**FWF-252913**	8.9
	29	17	14 800	25 500	1 510	2 600	17 000	**FWF-252917**	12
	30	13	13 500	19 500	1 380	1 980	17 000	**FWF-253013**	11
	30	17	17 200	26 400	1 750	2 690	17 000	**FWF-253017**	14
	30	20	18 800	29 700	1 920	3 050	17 000	**FWF-253020**	15
	30	25	23 600	39 500	2 410	4 050	17 000	**FWF-253025**	21
	31	13	15 000	19 500	1 520	1 990	17 000	**FWF-253113**	17
	31	17	19 200	26 800	1 950	2 740	17 000	**FWF-253117**	18
	31	20	22 100	32 000	2 250	3 300	17 000	**FWF-253120**	21
	31	24	26 500	40 500	2 700	4 150	17 000	**FWF-253124**	26
	32	16	18 700	23 500	1 910	2 390	17 000	**FWF-253216**	20
	32	24	28 300	40 000	2 880	4 050	17 000	**FWF-253224**	30
	33	20	28 400	37 000	2 900	3 750	18 000	**FWJ-253320**	30
	33	24	33 000	45 000	3 400	4 600	18 000	**FWJ-253324**	36
26	30	13	10 400	16 200	1 060	1 660	16 000	**FWF-263013**	9.3
	30	17	14 400	24 900	1 470	2 540	16 000	**FWF-263017**	12
	30	24	17 900	32 500	1 820	3 350	16 000	**FBN-263024**	20

FWF • FWJ

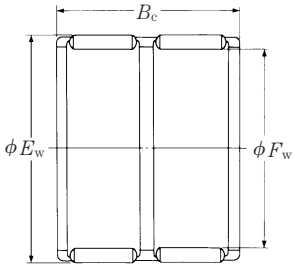
FWF···W

FBN···W

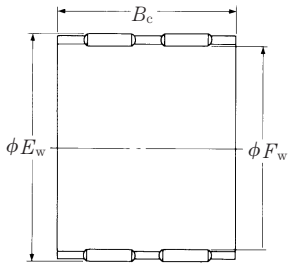
Inscribed Circle
Diameter 28 — 34 mm



FWF, FWJ



FWF···W



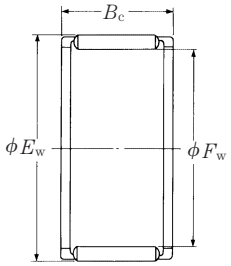
FBN···W

Boundary Dimensions (mm)			Basic Load Ratings (N)		Basic Load Ratings {kgf}		Limiting Speeds (min^{-1})	Bearing Numbers	Mass (g)
F_w	E_w	B_c $^{-0.2}_{-0.55}$	C_r	C_{or}	C_r	C_{or}	Oil		approx
28	33	13	13 700	20 400	1 400	2 080	15 000	**FWF-283313**	13
	33	17	17 100	27 100	1 740	2 760	15 000	**FWF-283317**	16
	33	20	21 900	37 500	2 240	3 800	15 000	**FWJ-283320**	20
	33	26	27 100	49 000	2 760	5 000	15 000	**FWF-283326**	25
	33	27	28 100	51 500	2 860	5 250	15 000	**FWF-283327**	26
	34	17	19 900	29 100	2 020	2 970	15 000	**FWF-283417**	20
	35	20	25 200	35 500	2 570	3 650	15 000	**FWF-283520**	22
	36	20	26 200	34 000	2 670	3 450	16 000	**FWF-283620**	33
29	34	17	17 500	28 400	1 790	2 900	14 000	**FWF-293417**	17
	34	24	25 400	46 000	2 590	4 650	14 000	**FWF-293424**	24
	34	27	26 600	48 500	2 710	4 950	14 000	**FWF-293427**	27
30	35	13	14 000	21 600	1 430	2 200	14 000	**FWF-303513**	14
	35	17	18 700	31 500	1 910	3 200	14 000	**FWF-303517A**	18
	35	20	21 000	36 000	2 140	3 700	14 000	**FWF-303520**	21
	35	27	28 700	54 000	2 920	5 500	14 000	**FWF-303527**	28
	36	14	18 400	26 900	1 870	2 750	14 000	**FWF-303614**	18
	36	20	24 300	38 500	2 480	3 950	14 000	**FWF-303620**	25
	36	26	29 600	49 500	3 000	5 050	14 000	**FWF-303626**	33
	37	16	22 200	31 000	2 270	3 150	14 000	**FWF-303716**	24
	37	20	26 000	38 000	2 650	3 850	14 000	**FWF-303720**	30
	38	18	25 500	34 000	2 600	3 450	14 000	**FWF-303818**	32
	38	20	25 900	34 500	2 640	3 500	14 000	**FWF-303820**	35
	38	27	38 500	57 500	3 950	5 850	14 000	**FWF-303827**	47
	40	25	38 500	49 500	3 950	5 050	15 000	**FWF-304025**	68
32	36	29	22 300	46 500	2 270	4 750	13 000	**FBN-323629WA**	32
	37	13	15 100	24 400	1 540	2 480	13 000	**FWF-323713**	14
	37	17	18 500	31 500	1 880	3 200	13 000	**FWJ-323717**	19
	37	23	24 300	44 500	2 480	4 550	13 000	**FWJ-323723**	25
	37	27	31 500	63 000	3 250	6 400	13 000	**FWJ-323727**	30
	37	28	25 200	47 000	2 570	4 800	13 000	**FWF-323728W**	31
	38	20	24 900	40 500	2 540	4 150	13 000	**FWF-323820**	27
	38	26	33 000	59 000	3 400	6 000	13 000	**FWF-323826**	35
	39	16	22 900	33 000	2 330	3 350	13 000	**FWJ-323916**	25
	39	20	27 300	41 000	2 780	4 200	13 000	**FWF-323920**	32
	40	20	28 400	39 500	2 890	4 000	13 000	**FWF-324020**	37
34	39	23	25 100	47 500	2 560	4 850	12 000	**FWJ-343923**	27

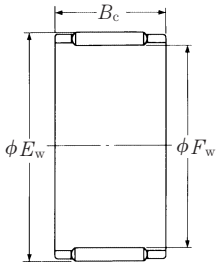
FWF • FWJ

FBN

Inscribed Circle
Diameter 35 — 38.1 mm



FWF, FWJ



FBN

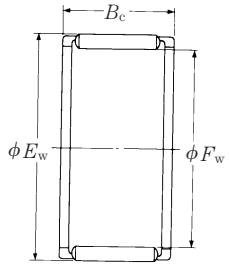
Boundary Dimensions (mm)			Basic Load Ratings (N)		Basic Load Ratings {kgf}		Limiting Speeds (min^{-1})	Bearing Numbers	Mass (g)
F_w	E_w	B_c $^{-0.2}_{-0.55}$	C_r	C_{or}	C_r	C_{or}	Oil		approx
35	40	13	14 900	24 600	1 520	2 500	12 000	**FWF-354013**	16
	40	17	20 500	37 000	2 090	3 750	12 000	**FWF-354017**	20
	40	20	26 400	51 000	2 690	5 200	12 000	**FWJ-354020**	26
	40	25	27 200	53 000	2 770	5 400	12 000	**FBN-354025**	30
	40	26	27 200	53 000	2 770	5 400	12 000	**FBN-354026**	31
	40	27	31 500	64 000	3 200	6 500	12 000	**FWF-354027**	32
	40	30	34 000	71 500	3 500	7 300	12 000	**FWF-354030**	34
	40	32.7	30 500	61 000	3 100	6 250	12 000	**FBN-354032Z**	40
	42	16	22 700	33 500	2 310	3 400	12 000	**FWF-354216**	28
	42	18	25 900	39 500	2 640	4 000	12 000	**FWF-354218**	31
	42	20	30 000	47 500	3 050	4 850	12 000	**FWJ-354220**	34
	42	30	43 000	75 500	4 350	7 700	12 000	**FWF-354230**	54
	44	18	29 600	39 500	3 000	4 050	12 000	**FWF-354418**	41
36	41	26	30 000	61 000	3 050	6 250	11 000	**FWF-364126**	31
	41	30.5	32 000	66 000	3 250	6 700	11 000	**FWJ-364130Z**	37
	48	22.5	45 500	57 500	4 650	5 900	12 000	**FWF-364822Z**	72
	48	25	48 500	62 500	4 950	6 350	12 000	**FWF-364825**	80
37	42	13	16 500	28 400	1 680	2 900	11 000	**FWF-374218**	16
	42	14.5	15 400	26 000	1 570	2 650	11 000	**FWJ-374214ZA**	15
	42	17	20 300	37 000	2 070	3 800	11 000	**FWJ-374217**	21
	42	27	31 500	66 000	3 250	6 700	11 000	**FWJ-374227**	34
	42	30	35 500	77 000	3 650	7 850	11 000	**FWF-374230**	43
	44	18	26 500	41 500	2 700	4 200	11 000	**FWF-374418**	32
38	43	13	16 400	28 500	1 670	2 900	11 000	**FWF-384313**	17
	43	17	20 400	38 000	2 080	3 850	11 000	**FWF-384317**	22
	43	27	32 500	68 500	3 300	7 000	11 000	**FWF-384327**	35
	44	27	39 000	76 500	4 000	7 800	11 000	**FWJ-384427**	42
	46	20	31 500	47 000	3 200	4 800	11 000	**FWF-384620**	43
	46	32	47 500	80 500	4 850	8 200	11 000	**FWF-384632**	75
38.1	50.8	24.7	50 500	64 000	5 150	6 500	11 000	**FWJ-385024Z**	93

FWF • FWJ

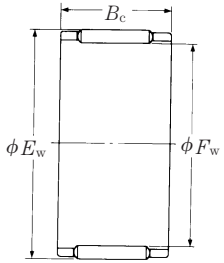
FWF···W

FBN

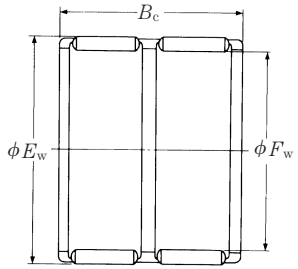
Inscribed Circle
Diameter 40 — 52 mm



FWF, FWJ



FBN



FWF···W

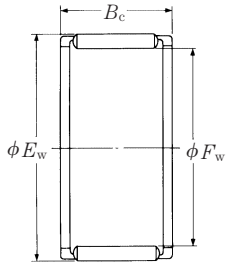
Boundary Dimensions (mm)			Basic Load Ratings (N)		Basic Load Ratings {kgf}		Limiting Speeds (min^{-1})	Bearing Numbers	Mass (g)
F_w	E_w	B_c $^{-0.2}_{-0.55}$	C_r	C_{or}	C_r	C_{or}	Oil		approx
40	45	13	15 700	27 500	1 610	2 800	10 000	**FWF-404513**	18
	45	17	21 000	40 000	2 150	4 050	10 000	**FWF-404517A**	23
	45	27	32 000	68 000	3 250	6 900	10 000	**FWF-404527**	36
	45	30	35 000	77 000	3 600	7 850	10 000	**FWJ-404530**	40
	46	30	35 000	67 000	3 550	6 800	10 000	**FWF-404630**	50
	48	20	32 500	49 500	3 300	5 050	10 000	**FWF-404820**	45
	48	25	40 500	66 500	4 150	6 800	10 000	**FWF-404825**	56
	48	34	53 500	95 000	5 450	9 700	10 000	**FWF-404834**	76
42	47	17	19 600	37 000	2 000	3 750	10 000	**FBN-424717**	24
	47	27	30 500	65 000	3 100	6 600	10 000	**FBN-424727**	38
	50	20	33 000	52 500	3 400	5 350	10 000	**FWF-425020**	47
43	48	17	22 800	45 500	2 330	4 650	9 500	**FWF-434817**	25
	48	27	33 500	75 000	3 450	7 650	9 500	**FWF-434827**	39
	49	31	44 000	92 000	4 450	9 400	9 500	**FWJ-434931**	55
45	50	17	21 600	43 000	2 200	4 350	9 000	**FWF-455017**	26
	50	27	34 000	77 500	3 500	7 900	9 000	**FWF-455027**	41
	51	35	46 000	99 500	4 700	10 100	9 000	**FWJ-455135**	61
	53	20	34 000	55 000	3 450	5 600	9 500	**FWF-455320**	50
	53	25	44 000	77 000	4 500	7 850	9 500	**FWF-455325**	62
46	52	37	43 500	94 000	4 450	9 600	9 000	**FWF-465237W**	75
47	52	17	23 400	48 500	2 390	4 900	8 500	**FWF-475217**	27
	52	27	34 500	79 500	3 500	8 150	8 500	**FWF-475227**	43
	52	30	36 500	86 000	3 750	8 750	8 500	**FWF-475230**	47
	55	28	51 500	96 000	5 250	9 800	9 000	**FWF-475528**	73
48	53	23.5	30 000	67 000	3 050	6 850	8 500	**FBN-485323Z**	47
	54	20	29 800	58 500	3 050	5 950	8 500	**FWF-485420**	39
50	55	20	26 900	59 000	2 750	6 050	8 000	**FWF-505520**	37
	55	27	35 000	83 000	3 600	8 450	8 000	**FWF-505527**	50
	55	30	39 500	96 500	4 000	9 800	8 000	**FWF-505530**	56
	56	20	28 100	54 500	2 870	5 600	8 000	**FBN-505620**	51
	58	20	38 500	67 500	3 950	6 900	8 500	**FWF-505820**	61
	58	25	48 500	90 500	4 950	9 200	8 500	**FWF-505825**	77
51	56	50.8	60 000	166 000	6 100	16 900	8 000	**FWF-515650WZ**	95
52	60	24	46 000	86 000	4 700	8 750	8 000	**FWF-526024**	46

FWF

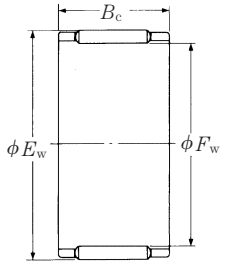
FWF···W

FBN

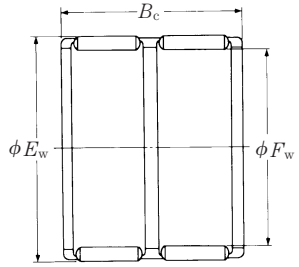
Inscribed Circle
Diameter 55 — 73 mm



FWF



FBN



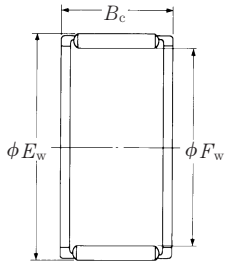
FWF···W

Boundary Dimensions (mm)			Basic Load Ratings (N)		Basic Load Ratings {kgf}		Limiting Speeds (min^{-1})	Bearing Numbers	Mass (g)
F_w	E_w	B_c $^{-0.2}_{-0.55}$	C_r	C_{or}	C_r	C_{or}	Oil		approx
55	60	20	27 500	62 500	2 800	6 350	7 500	**FWF-556020**	41
	60	27	36 000	89 000	3 700	9 100	7 500	**FWF-556027**	55
	60	30	41 000	105 000	4 200	10 700	7 500	**FWF-556030**	61
	61	20	31 000	64 000	3 150	6 500	7 500	**FWF-556120**	53
	61	30	47 000	109 000	4 750	11 100	7 500	**FWF-556130**	81
	63	20	40 000	73 000	4 100	7 450	7 500	**FWF-556320**	70
	63	25	50 000	97 500	5 100	9 950	7 500	**FWF-556325**	85
56	61	30	38 500	97 500	3 950	9 950	7 100	**FWF-566130**	62
	61	33.5	41 000	105 000	4 150	10 700	7 100	**FWF-566133Z**	70
57	73	30	89 500	129 000	9 150	13 100	7 500	**FWF-577330A**	77
58	64	40	54 000	132 000	5 500	13 500	7 100	**FWF-586440W**	104
	65	26.5	45 000	95 000	4 600	9 700	7 100	**FWF-586526Z**	81
60	65	20	29 000	69 000	2 960	7 050	6 700	**FWF-606520**	44
	65	27	38 000	98 500	3 900	10 000	6 700	**FWF-606527**	62
	65	41	50 500	141 000	5 150	14 400	6 700	**FWF-606541W**	91
	66	20	33 000	71 500	3 350	7 300	6 700	**FWF-606620**	57
	66	30	50 000	122 000	5 100	12 400	6 700	**FWF-606630**	87
	67	23	41 500	86 500	4 250	8 800	6 700	**FWF-606723**	73
	68	20	41 500	78 000	4 200	7 950	6 700	**FWF-606820**	73
	68	23	47 500	94 000	4 850	9 600	6 700	**FWF-606823**	84
	68	25	52 000	105 000	5 300	10 700	6 700	**FWF-606825**	91
	68	27	51 500	103 000	5 250	10 500	6 700	**FWF-606827**	98
65	70	20	27 900	67 500	2 850	6 900	6 300	**FBN-657020**	48
	70	30	44 500	123 000	4 550	12 500	6 300	**FWF-657030**	72
	73	30	61 000	132 000	6 200	13 400	6 300	**FWF-657330**	120
68	74	20	33 000	74 000	3 350	7 550	6 000	**FBN-687420**	60
	74	30	52 000	135 000	5 300	13 700	6 000	**FWF-687430**	91
70	76	20	35 500	82 500	3 600	8 450	5 600	**FWF-707620**	62
	76	30	53 500	141 000	5 450	14 400	5 600	**FWF-707630**	93
	78	30	63 000	140 000	6 400	14 300	6 000	**FWF-707830**	125
72	78	40	60 000	163 000	6 100	16 700	5 600	**FWF-727840W**	145
	79	21	42 000	93 000	4 300	9 500	5 600	**FWF-727921**	79
73	79	20	36 000	85 500	3 650	8 700	5 600	**FBN-737920**	65
	79	30	55 000	147 000	5 600	15 000	5 600	**FWF-737930**	97

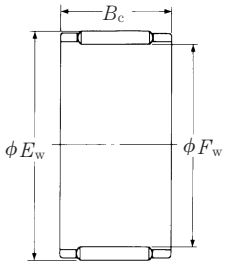
FWF

FBN

Inscribed Circle
Diameter 75 — 150 mm



FWF



FBN

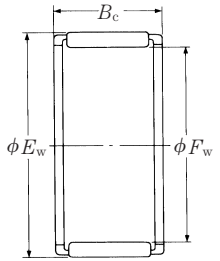
Boundary Dimensions (mm)			Basic Load Ratings (N)		Basic Load Ratings {kgf}		Limiting Speeds (min^{-1})	Bearing Numbers	Mass (g)
F_w	E_w	B_c $^{-0.2}_{-0.55}$	C_r	C_{or}	C_r	C_{or}	Oil		approx
75	81	20	37 000	89 500	3 750	9 100	5 300	**FBN-758120**	66
	81	30	54 500	147 000	5 550	15 000	5 300	**FWF-758130**	100
	83	30	65 000	151 000	6 650	15 400	5 600	**FWF-758330**	135
78	85	40	76 500	207 000	7 850	21 100	5 300	**FWF-788540**	160
80	86	20	38 000	94 000	3 850	9 600	5 000	**FWF-808620**	71
	86	30	57 000	160 000	5 850	16 300	5 000	**FWF-808630**	105
	88	30	69 000	166 000	7 050	17 000	5 000	**FWF-808830**	145
85	92	30	64 000	169 000	6 500	17 200	4 800	**FWF-859230**	130
	93	25	59 000	139 000	6 050	14 200	4 800	**FWF-859325**	125
	93	30	71 000	176 000	7 250	17 900	4 800	**FWF-859330**	150
90	98	30	70 000	177 000	7 150	18 000	4 500	**FWF-909830**	160
95	103	30	69 500	177 000	7 100	18 100	4 300	**FWF-9510330**	175
100	108	30	75 500	201 000	7 700	20 500	4 000	**FWF-10010830**	185
105	113	27	69 500	183 000	7 100	18 700	3 800	**FWF-10511327**	165
	113	30	72 500	193 000	7 400	19 700	3 800	**FWF-10511330**	185
110	118	30	78 500	218 000	8 000	22 300	3 600	**FWF-11011830**	195
115	123	35	92 500	274 000	9 450	28 000	3 400	**FWF-11512335**	235
120	128	25	69 500	192 000	7 100	19 600	3 400	**FWF-12012825**	190
125	133	35	93 500	284 000	9 550	29 000	3 200	**FWF-12513335**	255
130	138	25	71 500	205 000	7 300	20 900	3 000	**FWF-13013825**	220
135	143	38	107 000	345 000	10 900	35 500	3 000	**FWF-13514338**	275
145	153	30	89 000	281 000	9 050	28 700	2 800	**FWF-14515330**	290
150	160	43	149 000	475 000	15 200	48 000	2 600	**FWF-15016043**	505

WJ

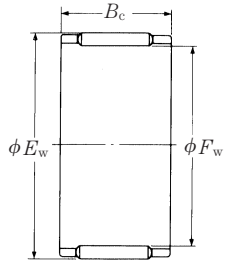
WJC

WJP

Inscribed Circle
Diameter 9.525 — 44.450 mm



WJ



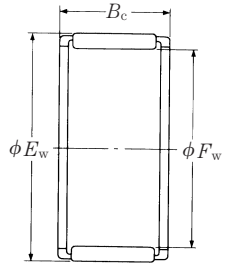
WJC, WJP

Boundary Dimensions (mm, inch) F_w		E_w		B_c $^{0}_{-0.38}$		Basic Load (N) C_r	C_{or}	Ratings {kgf} C_r	C_{or}	Limiting Speeds (min^{-1}) Oil	Bearing Numbers	Mass (g) approx
9.525	0.3750	12.700	0.5000	9.52	0.3750	3 900	4 000	400	410	45 000	**WJC-060806**	3
		12.700	0.5000	12.70	0.5000	5 450	6 150	555	625	45 000	**WJC-060808**	4
12.700	0.5000	15.875	0.6250	12.70	0.5000	6 350	8 100	645	825	34 000	**WJC-081008**	4
		15.875	0.6250	15.88	0.6250	8 000	10 900	815	1 110	34 000	**WJC-081010**	6
14.288	0.5625	17.462	0.6875	12.70	0.5000	6 900	9 350	705	955	30 000	**WJC-091108**	6
15.875	0.6250	19.050	0.7500	12.70	0.5000	7 100	10 000	725	1 020	26 000	**WJC-101208**	6
		22.225	0.8750	15.88	0.6250	15 400	17 300	1 570	1 770	28 000	**WJ-101410**	13
		22.225	0.8750	22.22	0.8750	21 100	25 900	2 150	2 640	28 000	**WJ-101414**	18
19.050	0.7500	25.400	1.0000	25.40	1.0000	26 600	36 500	2 710	3 750	22 000	**WJ-121616**	25
20.638	0.8125	26.988	1.0625	22.22	0.8750	24 600	33 500	2 510	3 450	20 000	**WJ-131714**	21
22.225	0.8750	28.575	1.1250	25.40	1.0000	29 000	42 500	2 960	4 350	20 000	**WJ-141816**	28
25.400	1.0000	33.338	1.3125	19.05	0.7500	27 800	36 000	2 830	3 700	17 000	**WJ-162112**	35
		33.338	1.3125	25.40	1.0000	36 500	51 500	3 700	5 250	17 000	**WJ-162116**	43
28.575	1.1250	38.100	1.5000	25.40	1.0000	42 000	56 500	4 300	5 750	15 000	**WJ-182416**	58
		38.100	1.5000	31.75	1.2500	51 500	73 500	5 250	7 500	15 000	**WJ-182420**	72
31.750	1.2500	41.275	1.6250	19.05	0.7500	33 000	42 500	3 350	4 350	14 000	**WJ-202612**	51
		41.275	1.6250	25.40	1.0000	44 000	61 000	4 450	6 250	14 000	**WJ-202616**	61
		41.275	1.6250	31.75	1.2500	53 500	79 500	5 500	8 100	14 000	**WJ-202620**	73
		41.275	1.6250	38.10	1.5000	63 000	98 000	6 450	10 000	14 000	**WJ-202624**	105
34.925	1.3750	44.450	1.7500	25.40	1.0000	45 500	66 000	4 650	6 700	12 000	**WJ-222816**	67
		44.450	1.7500	31.75	1.2500	55 500	85 500	5 700	8 750	12 000	**WJ-222820**	83
38.100	1.5000	47.625	1.8750	25.40	1.0000	47 000	70 500	4 800	7 200	11 000	**WJ-243016**	78
		47.625	1.8750	31.75	1.2500	57 500	91 500	5 850	9 350	11 000	**WJ-243020**	90
		47.625	1.8750	38.10	1.5000	68 000	113 000	6 900	11 500	11 000	**WJ-243024**	120
41.275	1.6250	57.150	2.2500	42.86	1.6875	96 500	130 000	9 850	13 300	11 000	***WJP-263627**	205
44.450	1.7500	53.975	2.1250	19.05	0.7500	39 000	58 000	4 000	5 950	9 500	**WJ-283412**	62
		53.975	2.1250	25.40	1.0000	51 500	83 500	5 250	8 500	9 500	**WJ-283416**	84
		53.975	2.1250	38.10	1.5000	74 500	134 000	7 600	13 700	9 500	**WJ-283424**	145

* These bearings have polyamide cages. The maximum permissible operating temperature for these bearings is 100°C for continued operation and 120°C for short periods.

WJ

Inscribed Circle Diameter 50.800 — 127.000 mm

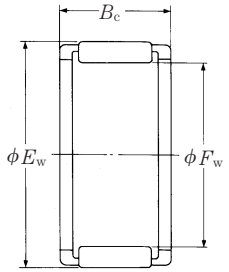


WJ

Boundary Dimensions (mm, inch) F_w		E_w		B_c $^{0}_{-0.38}$		Basic Load (N) C_r	C_{or}	Ratings {kgf} C_r	C_{or}	Limiting Speeds (min^{-1}) Oil	Bearing Numbers	Mass (g) approx
50.800	2.0000	60.325	2.3750	19.05	0.7500	42 500	67 500	4 300	6 850	8 500	**WJ-323812**	85
		60.325	2.3750	25.40	1.0000	56 000	96 500	5 700	9 850	8 500	**WJ-323816**	105
		60.325	2.3750	31.75	1.2500	68 500	126 000	7 000	12 800	8 500	**WJ-323820**	125
		60.325	2.3750	38.10	1.5000	81 000	155 000	8 250	15 800	8 500	**WJ-323824**	145
52.388	2.0625	61.912	2.4375	25.40	1.0000	57 500	101 000	5 850	10 300	8 000	**WJ-333916**	99
53.975	2.1250	63.500	2.5000	25.40	1.0000	53 000	92 000	5 400	9 400	8 000	**WJ-344016**	100
		63.500	2.5000	38.10	1.5000	78 000	151 000	7 950	15 400	8 000	**WJ-344024**	150
55.562	2.1875	65.088	2.5625	19.05	0.7500	44 000	73 500	4 500	7 500	7 500	**WJ-354112**	85
		65.088	2.5625	25.40	1.0000	58 500	105 000	5 950	10 700	7 500	**WJ-354116**	105
57.150	2.2500	66.675	2.6250	25.40	1.0000	53 500	94 000	5 450	9 600	7 500	**WJ-364216**	105
60.325	2.3750	69.850	2.7500	38.10	1.5000	81 000	165 000	8 300	16 800	7 100	**WJ-384424**	185
63.500	2.5000	73.025	2.8750	25.40	1.0000	55 500	102 000	5 650	10 400	6 700	**WJ-404616**	135
		73.025	2.8750	38.10	1.5000	82 500	171 000	8 450	17 500	6 700	**WJ-404624**	180
69.850	2.7500	79.375	3.1250	25.40	1.0000	57 500	110 000	5 850	11 200	6 000	**WJ-445016**	125
76.200	3.0000	85.725	3.3750	25.40	1.0000	59 000	118 000	6 050	12 000	5 300	**WJ-485416**	160
		85.725	3.3750	38.10	1.5000	85 000	189 000	8 700	19 200	5 300	**WJ-485424**	215
82.550	3.2500	92.075	3.6250	25.40	1.0000	61 000	126 000	6 200	12 900	5 000	**WJ-525816**	145
		92.075	3.6250	38.10	1.5000	87 500	201 000	8 950	20 500	5 000	**WJ-525824**	220
88.900	3.5000	98.425	3.8750	25.40	1.0000	62 500	134 000	6 400	13 700	4 500	**WJ-566216**	175
		101.600	4.0000	25.40	1.0000	79 000	147 000	8 050	15 000	4 500	**WJ-566416**	215
		101.600	4.0000	38.10	1.5000	113 000	234 000	11 500	23 800	4 500	**WJ-566424**	360
101.600	4.0000	114.300	4.5000	25.40	1.0000	83 000	163 000	8 450	16 600	4 000	**WJ-647216**	240
		114.300	4.5000	38.10	1.5000	119 000	259 000	12 100	26 400	4 000	**WJ-647224**	370
127.000	5.0000	152.400	6.0000	38.10	1.5000	211 000	360 000	21 500	36 500	3 400	**WJ-809624**	1 000

For Large Ends of Connecting Rods

Inscribed Circle Diameter 12 — 30 mm

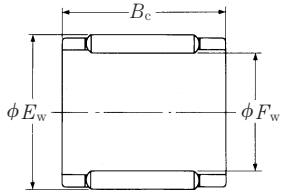


FWF···E

Boundary Dimensions (mm)			Basic Load Ratings (N)		{kgf}		Bearing Numbers	Mass (g)
F_w	E_w	B_c $^{-0.2}_{-0.4}$	C_r	C_{or}	C_r	C_{or}		approx
12	16	10	6 100	6 500	620	665	**FWF-121610-E**	4.0
14	19	10	7 800	8 050	795	820	**FWF-141910-E**	6.2
	20	12	8 900	8 600	910	880	**FWF-142012-E**	8.3
15	19	9	5 650	6 250	575	640	**FWF-15199-E**	4.1
	20	10	7 300	7 600	745	775	**FWF-152010-E**	6.0
	21	10	7 950	7 500	810	765	**FWF-152110-E**	8.5
16	21	11	8 650	9 600	880	980	**FWF-162111-E**	7.5
	22	12	9 500	9 600	965	980	**FWF-162212-E**	9.5
18	23	14	11 800	14 800	1 200	1 510	**FWF-182314-E**	10
	24	12	10 000	10 600	1 020	1 080	**FWF-182412-E**	11
20	26	12	12 200	14 100	1 250	1 440	**FWF-202612-E**	13
	26	17	16 800	21 200	1 710	2 160	**FWF-202617-E**	17
	28	18	18 100	19 400	1 840	1 970	**FWF-202818-E**	25
22	28	14	13 900	17 100	1 420	1 740	**FWF-222814-E**	14
	29	15	16 300	19 000	1 660	1 930	**FWF-222915-E**	19
	32	16	19 700	19 400	2 010	1 970	**FWF-223216-E**	31
23	31	16	17 600	19 400	1 800	1 980	**FWF-233116-E**	23
24	30	15	15 600	20 300	1 590	2 070	**FWF-243015-E**	17
	30	17	17 900	24 300	1 830	2 480	**FWF-243017-E**	19
	31	20	21 600	27 800	2 200	2 840	**FWF-243120-E**	30
25	32	16	17 700	21 900	1 810	2 230	**FWF-253216-E**	24
28	35	16	18 400	23 700	1 880	2 410	**FWF-283516-E**	25
29.75	36.75	16.5	19 600	26 000	1 990	2 650	**FWF-293616Z-E**	28
30	37	16	21 900	30 500	2 230	3 100	**FWF-303716-E**	29
	38	18	25 500	34 000	2 600	3 450	**FWF-303818-E**	35

For Small Ends of Connecting Rods

Inscribed Circle Diameter 9 — 19 mm



FBN···E

Boundary Dimensions (mm)			Basic Load Ratings (N)		{kgf}		Bearing Numbers	Mass (g)
F_w	E_w	B_c $^{-0.2}_{-0.4}$	C_r	C_{or}	C_r	C_{or}		approx
9	12	11.5	4 200	4 500	430	460	**FBN-91211Z-E**	3.5
10	14	12.7	5 900	5 950	605	610	**FBN-101412Z-E**	5.0
12	15	14.3	6 400	8 400	655	855	**FBN-121514Z-E**	4.8
	16	13	7 250	8 200	740	835	**FBN-121613-E**	6.4
	16	15.5	8 500	10 000	865	1 020	**FBN-121615Z-E**	7.0
	16	16	8 500	10 000	865	1 020	**FBN-121616-E**	7.5
14	18	12	6 950	8 050	710	820	**FBN-141812-E**	6.5
	18	16.5	9 250	11 600	945	1 180	**FBN-141816Z-E**	8.5
	18	18	10 700	14 000	1 090	1 430	**FBN-141818-E**	11.5
	18	20	9 550	12 000	975	1 230	**FBN-141820-E1**	13
15	19	18	11 300	15 300	1 150	1 560	**FBN-151918-E**	11
	21	18	12 900	13 900	1 310	1 420	**FBN-152118-E**	13
16	20	22	13 700	20 000	1 400	2 040	**FBN-162022-E**	14
	20	23.5	14 900	22 300	1 520	2 280	**FBN-162023Z-E**	15
	21	20	14 200	18 100	1 450	1 840	**FBN-162120-E**	16
17	21	23	14 800	22 500	1 510	2 290	**FBN-172123-E**	16
18	22	17	11 500	16 500	1 170	1 680	**FBN-182217-E**	12
	22	22	14 200	21 600	1 440	2 200	**FBN-182222-E**	15
	22	23.6	15 400	24 100	1 570	2 460	**FBN-182223Z-E**	16
19	23	23.7	16 000	25 800	1 630	2 630	**FBN-192323Z-E**	17

Please consult NSK when making a selection.



DRAWN CUP NEEDLE ROLLER BEARINGS

METRIC DESIGN

Pages

Open Type with Cage and Closed End Type with Cage
Inscribed Circle Diameter 10–55 mm B46–B53
Sealed Type with Cage — Inscribed Circle Diameter 8–50 mm B54–B57
Thin, Open Type with Cage — Inscribed Circle Diameter 18–82 mm B58–B59
Full Complement Open Type and Full Complement Closed End Type
Inscribed Circle Diameter 8–55 mm B60–B63

INCH DESIGN

Open Type with Cage and Closed End Type with Cage
Inscribed Circle Diameter 9.525–50.800 mm B64–B67
Full Complement Open Type and Full Complement Closed End Type
Inscribed Circle Diameter 11.112–50.800 mm B68–B75
Full Complement and Grease-Retained Roller Type
Inscribed Circle Diameter 3.969–47.625 mm B76–B79

EXTRA SMALL · MINIATURE

Metric Design Open Type with Cage and Closed End Type with Cage
Inscribed Circle Diameter 3–9 mm B80–B81
Full Complement Metric Design Open Type and Full Complement Closed End Type
Inscribed Circle Diameter 3.036–9 mm B82–B83
Inch Design Open Type with Cage and Closed End Type with Cage
Inscribed Circle Diameter 3.175–9.525 mm B84–B85
Inch Design Full Complement Open Type and Full Complement Closed End Type
Inscribed Circle Diameter 3.175–9.525 mm B86–B87

NSK drawn cup needle roller bearings (with a cage and full complement of rollers) have a unique outer ring structure. They have the following features: the thinnest outer ring among rolling bearings, high load capacity, high maximum permissive load owing to the use of carefully selected special alloy steel plates, a surface-hardened cage with high resistance to wear, high limiting speed, and ease of mounting.

DESIGN AND TYPES

NSK drawn cup needle roller bearings consist of an outer ring made of carefully selected alloy steel that is surface-hardened after precise pressing, rollers made of vacuum-degassed bearing steel that are given a grinding finish after through-hardening, and a cage that guides the rollers correctly on the pitch diameter.

The bent lip of the drawn cup, which is the outer ring, improves rigidity, holds the rollers and cage in place, and functions as a labyrinth seal to prevent foreign matter from entering and lubrication from leaking. The inside of the outer ring's bending part is hardened, and has sufficient wear resistance against the contact of the roller or cage.

The bearing numbers of NSK drawn cup bearings are shown in combination with bearing codes and dimension numbers.
The bearing codes are shown in **Table 1**.

Table 1 Bearing Codes

Items		With Cage Assembly		Full Complement	
		Open	Closed End	Open	Closed End
Standard Series	Metric	FJ, FJP	MFJ	F, NF, DD	MF, MNF
	Inch	J, JP	MJ	B, Y	M
High Load Series	Metric	FJL, FJH	MFJL, MFJH	FH, NFH	MFH
	Inch	JH	MJH	BH, YH	MH

The dimension number has the following meaning:

Metric	The first one or two digits indicate the shaft's dimensions, and the following digits indicate the width, both expressed in mm units.
Inch	The first one or two digits indicate the shaft's dimensions, and the following digits indicate the width, both expressed in units of 1/16 of an inch. With closed ends, the number "1" is added following the dimension number.

(Example) FJ-1612-OH

- OH — Special Specification Code
- 12 — Width Dimension
- 16 — Shaft Dimension
- (12, 16) — Dimension Number
- FJ — Code

FJ means a metric open-type bearing with a cage.

Special Specifications

The bearing is manufactured in accordance with special specifications depending on operating conditions. In this case, the code is indicated with the bearing number followed by a hyphen and a letter or alphanumeric combination.

(Example) OH .... outer ring with oil hole

ACCURACY

Although standard drawn cup needle bearings perform well for general use, higher-accuracy drawn cup needle bearings are sometimes required such as when dispersion of radial internal clearance needs to be sharply decreased, when eccentric effects should be minimized, or when using only a few bearings to share the load evenly. However, standard bearings are still sufficient for almost all other occasions. Considering the higher cost of high-accuracy bearings, please consult NSK when selecting such bearings.

INTERNAL CLEARANCE

The radial internal clearance when a drawn cup needle bearing is mounted in a housing varies depending on the bearing, shaft and housing bore tolerances. Therefore, the tolerances for the shaft and housing bore should be narrower when reducing the range of radial internal clearance.

When the dimensional tolerance for each part is as shown in **Fig. 1** (Example of **F-3020-1**), the radial internal clearance ranges from 0.015 to 0.074 mm.

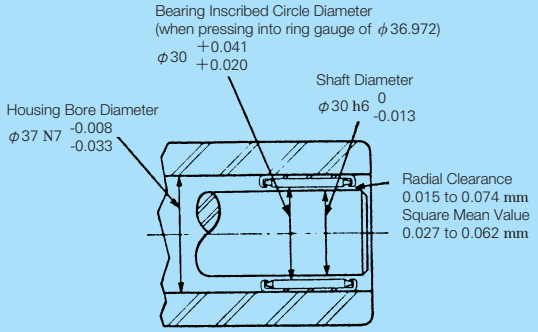


Fig. 1 Radial Internal Clearance

APPLICATION

Shaft and housing specifications

For drawn cup needle bearings, the shaft is generally used as a substitute for the inner ring. Since the outer ring of drawn cup needle bearings is made of a thin steel plate, the correct shape and dimensional accuracy are achieved after press fitting the regular housing. The dimensional accuracy and hardness of the shaft and housing bore directly affect bearing performance. Therefore, the following specifications should be satisfied.

(1) Accuracy, Roughness and Hardness

Accuracy, roughness and hardness of the shaft and housing bore should satisfy the specifications shown in **Table 2**.

(2) Shaft Inclination

Shaft inclination due to deflection by external force and mounting error of the bearing should be within the values found in **Table 3**.

Table 2 Accuracy, Roughness, and Hardness

Category	Shaft	Housing Bore
Out-of-Roundness Tolerance	$\frac{IT3}{2}$	$\frac{IT4}{2}$ to $\frac{IT5}{2}$
Cylindricity Tolerance	$\frac{IT3}{2}$	$\frac{IT4}{2}$ to $\frac{IT5}{2}$
Roughness, Ra	0.4	1.6
Hardness	HRC58 to 64 (HRC60 to 64 is favorable) Proper depth of hardening layer required	—

Remarks For details, refer to Section **5.2** on page A34 and Section **5.3** on page A35.

Table 3 Permissible Shaft Inclination

Bearing Width (mm)		Permissible Inclination (mm/mm)	
over	incl	With Cage	Full Complement
	25	0.0015	0.0010
25	50	0.0010	0.0005
50		0.0005	0.0005

(3) Housing Rigidity

To maintain the accuracy of drawn cup needle bearings, it is necessary to design the housing thick enough to minimize deformation when load is applied.

Split housings can only be used with drawn cup needle bearings after press fitting the bearing into a cylindrical sleeve.

FITTING

Drawn cup needle bearings perform as designed only after proper fitting. It is necessary to use the fitting tolerances shown in **Table 4** for the shaft and housing bore.

The figures in **Table 4** are for fits when housing materials are rigid steel or cast iron.

The fit should be adjusted as follows in accordance with operating conditions:

(1) Oscillating motion

In order to reduce radial internal clearance, the shaft's fitting tolerance should be js6 (j6).

(2) Outer ring rotation

For metric bearings, the fitting tolerances of the shaft and housing bore should be f6 and R7, respectively.

For inch bearings, 0.013 mm should be reduced from both fitting tolerances of shaft and housing bore.

(3) The housing is made from light alloy or steel with a thickness of less than 6 mm.

The fitting tolerances of housing bore should be minimized to approximately 0.013 to 0.025 mm, provided that the optimized value should be determined by mounting into the actual housing because the value may vary depending on housing material and shape. Please contact NSK for details.

Table 4 Recommended Fits

Category		Fitting Tolerance			
		Shaft	Housing Bore		
Metric	F, FJ	h6	N7		
	FJL	h6	J7		
Inch	B, BH, J, JH, Y, YH	h6	Nominal Outside Diameter D, mm		Tolerance, mm
			over	incl	high / low
				7.144 (9/32)	+0.013 / 0
			7.144 (9/32)	8.731 (11/32)	0 / +0.013
			8.731 (11/32)		J7 / J7

MOUNTING

When an drawn cup needle roller bearing is pressed into a housing bore, it is necessary to use an appropriate jig to prevent deformation of and damage to the outer ring, as shown in **Fig. 2**. The following precautions apply when mounting:

(1) The marked-side face should be placed onto the shoulder of the jig.

(2) A hand press should be used for press fitting. Avoid using a hammer or other such instruments.

(3) No snap ring or shoulder to determine the position of the bearing is necessary. However, when using a housing with a shoulder, the bearing side face should not touch the shoulder, in order to prevent bearing deformation.

(4) The guide and locaing stopper (**Fig. 2**) are installed in order to apply press fitting accurately. The ball detent shown in **Fig. 2** helps the jig hold the bearing and prevent roller inclination while facilitating mounting work.

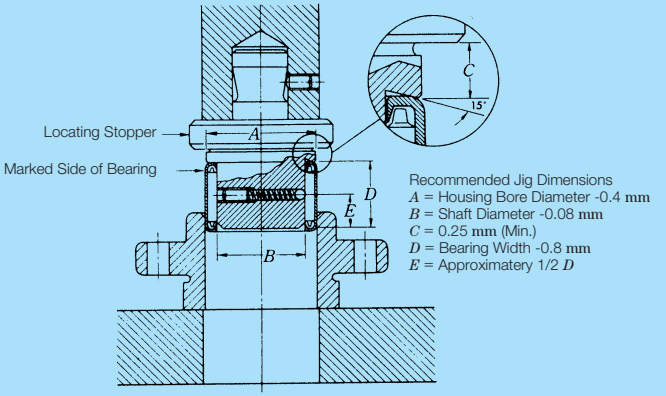


Fig. 2 Example of Jig

Inspection of Drawn Cup Needle Bearings

Deformation to a certain degree due to heat treatment is inevitable even though the outer ring of drawn cup needle bearings is processed by precise deep drawing. However, once the bearing is pressed into the regular housing, this deformation is corrected and the bearing performs to the full potential. Therefore, it is not appropriate to measure the drawn cup outside diameter before mounting. Correct inspection is carried out according to following procedures:

(1) The bearing that is to be subject to the test is pressed into a ring gauge of given dimensions.

(2) The inscribed circle diameter at the GO and NO-GO plug gauges is inspected. Metric inspection gauge dimensions are shown in **Table 5**.

Table 6 shows tolerance for the roller inscribed circle diameter based on ISO standards.

For bearings based on ISO standards, "-1" is added at the end of the bearing number. **Table 7** shows inch inspection gauge dimensions.

Table 5 Inspection Gauge Dimensions (General Metric)

Units: mm

Nominal Roller Inscribed Circle Diameter, F_w	Bore Diameter of Ring Gauge		Plug Gauge	
	F, MF, FH, MFH, NF, MNF, NFH, FJ, MFJ	FJL, MFJL	GO Gauge	NO-GO Gauge
4	7.996	—	4.023	4.048
5	8.996	—	5.023	5.048
6	9.996	—	6.028	6.053
7	10.995	—	7.031	7.056
8	11.995	15.010	8.031	8.056
9	12.995	16.010	9.031	9.056
10	13.995	17.010	10.031	10.056
12	15.995	19.012	12.031	12.056
FH12	17.995	—	12.031	12.056
13	18.993	—	13.034	13.059
14	19.993	22.012	14.034	14.059
NF14	18.993	—	14.034	14.059
15	20.993	22.012	15.034	15.059
16	21.993	24.012	16.034	16.059
17	22.972	23.991	17.013	17.038
18	23.972	24.991	18.013	18.038
20	25.972	26.991	20.013	20.038
22	27.972	28.991	22.013	22.038
NFH22	30.987	—	24.015	24.035
24	—	30.989	24.013	24.038
NFH24	31.967	—	24.013	24.038
25	31.967	32.989	25.013	25.038
28	34.967	36.989	28.013	28.038
30	36.967	39.989	30.013	30.038
32	—	41.989	32.013	32.038
35	41.967	44.989	35.013	35.043
40	46.967	49.989	40.013	40.043
45	51.961	54.988	45.013	45.043
50	57.961	61.988	50.013	50.043
55	62.961	66.988	55.013	55.043

Remarks This is the gauge dimension for inspection of minimum diameter, F_w, of roller inscribed circle diameter.

Table 6 Ring Gauge of Drawn Cup Needle Roller Bearings and Tolerance of Roller Inscribed Circle Diameter (ISO Standerds)

[FJ, FJH, MFJ and MFJH
F, FH, MF and MFH]

Units: mm

Nominal Roller Inscribed Circle Diameter, F_w	Bore Diameter of Ring Gauge	Tolerance for Roller Inscribed Circle Diameter, F_w(1)	
		min	max
3	6.484	3.007	3.021
4	7.984	4.010	4.028
5	8.984	5.010	5.028
6	9.984	6.010	6.028
7	10.980	7.013	7.031
8	11.980	8.013	8.031
H 8	13.980	8.013	8.031
9	12.980	9.013	9.031
H 9	14.980	9.013	9.031
10	13.980	10.013	10.031
H10	15.980	10.013	10.031
12	15.980	12.016	12.034
H12	17.980	12.016	12.034
13	18.976	13.016	13.034
14	19.976	14.016	14.034
15	20.976	15.016	15.034
16	21.976	16.016	16.034
17	22.976	17.016	17.034
18	23.976	18.016	18.034
20	25.976	20.020	20.041
22	27.976	22.020	22.041
25	31.972	25.020	25.041
28	34.972	28.020	28.041
30	36.972	30.020	30.041
35	41.972	35.025	35.050
40	46.972	40.025	40.050
45	51.967	45.025	45.050
50	57.967	50.025	50.050
55	62.967	55.030	55.060

Note (1) When using a cylinder instead of an inner ring, $F_{w\ min}$ is the diameter of the cylinder at which the internal clearance is zero in at least one radial direction. ($F_{w\ min}$ is the minimum diameter of each inscribed circle diameter where deviation is suspected.)

Remarks To measure the roller inscribed circle diameter, use the followung plug gauges:
GO gauge: The same dimensions as the minimum tolerance of the roller inscribed circle diameter, F_w
NO-GO gauge: The dimensions should be the maximum tolerance of roller inscribed circle diameter, F_w, plus 0.002 mm

Table 7 Inspection Gauge Dimensional Table (Inch)

B, M, J, MJ, Y and JP

Units: mm

Nominal Roller Inscribed Circle Diameter, F_w	Bore Diameter of Ring Gauge	Plug Gauge	
		GO Gauge	NO-GO Gauge
3.175(1/8)	6.363	3.195	3.220
3.969(5/32)	7.155	3.990	4.015
4.762(3/16)	8.730	4.783	4.808
6.350(1/4)	11.125	6.388	6.413
7.938(5/16)	12.713	7.976	8.001
9.525(3/8)	14.300	9.563	9.588
11.112(7/16)	15.888	11.151	11.176
12.700(1/2)	17.475	12.738	12.763
14.288(9/16)	19.063	14.326	14.351
15.875(5/8)	20.650	15.913	15.938
17.462(11/16)	22.238	17.501	17.526
19.050(3/4)	25.387	19.063	19.088
20.638(13/16)	26.975	20.650	20.675
22.225(7/8)	28.562	22.238	22.263
23.812(15/16)	30.150	23.825	23.850
25.400(1)	31.737	25.413	25.438
26.988(1 1/16)	33.325	27.000	27.025
28.575(1 1/8)	34.912	28.588	28.613
30.162(1 3/16)	38.087	30.175	30.200
31.750(1 1/4)	38.087	31.763	31.788
33.338(1 5/16)	41.262	33.350	33.378
34.925(1 3/8)	41.262	34.938	34.966
38.100(1 1/2)	47.612	38.113	38.143
41.275(1 5/8)	50.787	41.288	41.318
44.450(1 3/4)	53.962	44.463	44.496
47.625(1 7/8)	57.137	47.638	47.671
50.800(2)	60.312	50.815	50.848

BH, MH, JH, MJH and YH

Units: mm

Nominal Roller Inscribed Circle Diameter, F_w	Bore Diameter of Ring Gauge	Plug Gauge	
		GO Gauge	NO-GO Gauge
7.938(5/16)	14.300	7.976	8.001
9.525(3/8)	15.888	9.563	9.588
11.112(7/16)	17.475	11.151	11.176
12.700(1/2)	19.063	12.738	12.763
14.288(9/16)	20.650	14.326	14.351
15.875(5/8)	22.238	15.913	15.938
17.462(11/16)	23.825	17.501	17.526
19.050(3/4)	26.975	19.063	19.088
20.638(13/16)	28.562	20.650	20.675
22.225(7/8)	30.150	22.238	22.263
25.400(1)	33.325	25.413	25.438
28.575(1 1/8)	38.087	28.588	28.613
31.750(1 1/4)	41.262	31.763	31.788
34.925(1 3/8)	44.437	34.938	34.966

Drawn Cup Needle Bearings with Seals

Drawn cup needle roller-bearing with seals can be sealed on one or both sides. The bearings are packed with lithium soap-based grease and can be operated under regular conditions in normal operating ranges for long periods without replenishment. The seale made of nitrile synthetic rubber reinforced with a steel plate and serves as a guide for the side face of the cage. The operating temperature range is limited because of the rubber seal, ranging from -10°C to +100°C for continuous use from -25°C to +120°C for short periods of use.

The various types of drawn cup needle bearings with seals are shown in **Table 8**.

Table 8 Types of Drawn Cup Needle Roller Bearings with Seals

Code	Description
FJT, FJLT, FJHT	Drawn cup with seal on one side
FJTT, FJLTT, FJHTT	Drawn cup with seals on both sides
MFJT, MFJLT, MFJHT	Closed end, drawn cup with seal on one side